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2025-01-31 9:16:53 AM CST
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Thundercomm Turbox™ C410/C610 Open Kit Linux Android

Software Release Notes

Rev. V1.5
July 06, 2021

DN: tc-gla-2415
SDK Version: LA1.0.ES.r000002

Revision History

Revision	Date	Description
1.0	Oct 26, 2020	Initial release
1.1	Feb 04, 2021	Update Flash images to board.
1.2	Apr. 01, 2021	Update Chapter 1. Introduction.
1.3	May 26, 2021	Update the Step 1 in 3.3.1.Linux package dependencies.
1.4	May 31, 2021	Update the code downloading method of mainland China in 4.4.2.Download LA SDK code.
1.5	July 06, 2021	Update the SDK code download command in 4.4.2.Download LA SDK code.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of your computer, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders, path and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document aims to guide users to obtain, build, and program QCS610 Linux Android (LA) SDK into the Turbox™ C410/C610 Platform.

The Linux Android Software Open Kit supports the Android operating system on the Thundercomm Turbox™ SOM. The target of the LA SDK is to provide a simple and flexible development platform to rapidly build IoT products with the QTI chipsets. The LA SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

The LA SDK is based on AOSP (Android Open-Source Project) Android 10 release and follows several of its development principles.

➲ **NOTE:** The Linux Android Software can only run on 4G DDR Turbox™ C610 Platform.

Table 1-1. Turbox™ C610 patform secifications



Type	SOM	DDR Size	eMMC Size
SKU1	C610	2 GB	16 GB
SKU2	C610	2 GB	64 GB
SKU3	C610	4 GB	64 GB
SKU4	C410	2 GB	16 GB
SKU5	C410	2 GB	64 GB
SKU6	C410	1 GB	4 GB
SKU7	C410	4 GB	64 GB
SKU8	C410	2 GB	8 GB

Figure 1-1. SOM ID Location

Chapter 2. System Requirements

Table 2-1. Hardware and software required to install and run the LA SDK

Hardware / Software Requirements	Source / Vendor	Function
Standalone hardware system minimum requirement: 16 GB RAM Quadcore CPU ➲ **NOTE**: System with lower hardware specifications will increase compilation time. About an hour is the compilation time for a system with 16 GB RAM and Intel i7-2600 @3.4 GHz.	–	SDK build machine
Ubuntu 16.04 LTS Linux distribution for 64 bit architecture	Ubuntu Community/ Canonical, Ltd.	SDK build host OS
Repository	SDK Open Source Project	SDK source management tool
Python 2.7.12 (or a later version) with sqlite3 module	Python.org	Building subsystem

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Chapter 3. Development Environment Setup

This chapter introduces the setup of a Linux development environment on an Ubuntu host machine.

Prerequisites:

Ensure your host meets the following requirements:

- Host machine Linux distro – 16.04 LTS
- Minimum 300 GB disk free space
- **Tar**: V1.27 or newer
- Privileges to install utilities
- Install the host packages

3.1. Ubuntu 16.04 (64-bit) installation

Step 1.Install ubuntu 16.04 on the computer following the instructions at:
http://www.ubuntu.com/download/desktop/install-ubuntu-desktop.

Step 2.To update software, execute the following command:

```
$ sudo apt-get update
```

3.2. Tools and USB driver installation

This section provides instructions for installing **ADB** (Android Debug Bridge) (V1.0.36 or later), **Fastboot**, and USB drivers to your host computer.

The **Fastboot** is a tool used to flash images onto the board by communicating with the boot loader.

The **ADB** is a tool installed on the host computer to access the device Unix shell, then help to push, modify, debug, and tweak device system files from the host computer.

3.2.1. ADB and Fastboot tools installation

To install **ADB** and **Fastboot**, execute the following commands:

```
$ sudo apt-get install android-tools-adb
$ sudo apt-get install android-tools-fastboot
```

3.2.2. USB drivers installation ad configuration

To install and configure USB drivers, follow the steps below:

Step 1.Set up the udev rules on a Linux computer.

1. Login as **root/sudo**.

   ```
   $ cd /
   ```

2. Navigate to the directory */etc/udev/rules.d/*.

   ```
   $ cd /etc/udev/rules.d/
   ```

3. Create the file *99-android.rules.*

   ```
   $ sudo touch 99-android.rules
   ```

4. Enter the following command to get the write permission to file *99-android.rules*.

   ```
   $ sudo chmod 777 /etc/udev/rules.d/99-android.rules
   ```

5. Open the *99-android.rules* by running the following command.

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```
$ vim 99-android.rules
```

6. Input `i` to enter edit mode and add the following content to the file.

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", MODE="0777", GROUP="adm"
```

7. Press **ESC** key to exit the edit mode.
8. Enter `:wq` to save the change and exit.

Step 2. Restart the udev service by executing the following commands.

```
$ sudo chmod a+r /etc/udev/rules.d/99-android.rules
$ sudo service udev restart
```

Step 3. Connect the board to your computer with a USB cable, and power on the target board to start the system. Wait for several seconds then input `$ adb devices`. The device is correctly detected with the following information displayed on your screen.

```
$ adb devices
List of devices attached
7050372c        device
```

➲ **NOTE:** Avoid to connect the device to the PC with a USB hub. Connect the device to the PC directly via USB cable.

3.3. Required building environment

3.3.1. Java 8

Step 1. Check the version. Run the following commands to ensure Java 8 is installed.

```
$ java -version
openjdk version "1.8.0_131"
OpenJDK Runtime Environment (build 1.8.0_131-8u131-b11-0ubuntu1.16.04.2-b11)
OpenJDK 64-Bit Server VM (build 25.131-b11, mixed mode)
```

Step 2. Version update

With a 1.7 version preinstalled, run the following commands to upgrade it to version 1.8.

```
$ sudo apt-get update
$ sudo apt-get install openjdk-8-jdk
```

➲ **NOTE:** The Java 8 should be used along with Ubuntu 16.04.

3.3.2. Linux package dependencies

Step 1. Install the dependency libraries to the host computer:

```
$ sudo apt-get install git-core gnupg flex bison build-essential zip curl zlib1g-dev gcc-multilib g++-multilib libc6-dev-i386 lib32ncurses5-dev x11proto-core-dev libx11-dev lib32z1-dev libgl1-mesa-dev libxml2-utils xsltproc unzip fontconfig libxml-simple-perl
```

Step 2. Set **bash** as default shell tool:

```
$ sudo rm /bin/sh
$ sudo ln -sf /bin/bash /bin/sh
```

➲ **NOTE**: For more information on building environment dependencies, Refer to AOSP documents at: *https://source.android.com/setup/build/initializing*.

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3.3.3. Install repo

Run the following command to install repo.

```
$ sudo apt-get install repo
```

3.3.4. Configure git global variable

Configure user name and email:

```
$ git config --global user.email "<User Email>"
$ git config --global user.name "<User Name>"
```

➲ **NOTE:** Replace the `<User Email>` and `<User Name>` with the email address and user name you have registered at Thundercomm website.

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Chapter 4. Get Started with LA SDK

4.1. LA SDK information

Table 4-1. LA SDK information

Product	Baseline Distribution	SDK Version	OS Version
QCS610.LA.1.0	ES_0.0.010.2	LA1.0.ES.r000002	Android 10.00.00

4.2. LA SDK block diagram

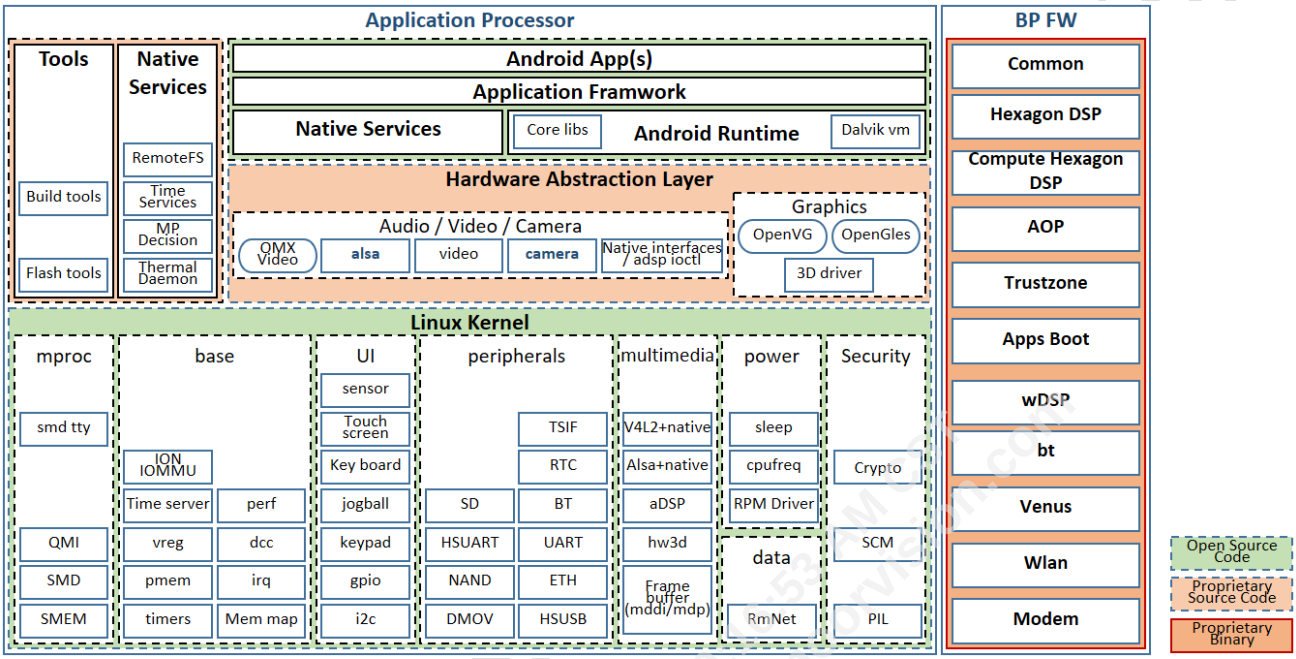


Figure 4-1. LA SDK Block Diagram

4.3. SDK source tree structure

```
SDK.Turbox_C610.LA1.0/
├── about.html -> common/about.html          SDK version info(Product,
Distribution,Components, …)
├── contents.xml -> common/contents.xml             image contents file
├── common/config/emmc/partition.xml                       eMMC partition file
├── LINUX
│   └── android
│       ├── bionic
│       ├── bootable/bootloader/edk2                bootloader sources
│       ├── bootable/recovery                       recovery sources
│       ├── build           basic config files for development kit, such as files for
system build rules and generis
│       ├── dalvik
│       ├── developers                   developer directory
│       ├── development              infrastructure-related files for Android application
development
│       └── device                   device related config
│           └── qcom/sm6150               QCS610 device-related config directory
│       ├── external                 files related to open-source development kit
│       ├── frameworks                   source code under Android frameworks
```

```
|       ├── hardware/qcom            code for QCS610 hardware abstraction layer (bootctrl,
Display, gps, media, vr, …)
|       ├── kernel
|       |   └── msm-4.14                                  kernel sources
|       |       ├── arch/arm64/boot/dts/qcom               DT source files
|       |       └──arch/arm64/configs/vendor               kernel config files
|       ├── libcore                                       core library files
|       ├── libnativehelper                                 dynamic library, which is the basis
to JNI library
|       ├── Makefile                                      global Makefile to define build rules
|       ├── packages                                      application packages
|       ├── pdk                                           plug development kit
|       ├── platform_testing                                platform testing files
|       ├── prebuilts                                       prebuild resources under x86 and arm
framework
|       ├── sdk                                           sdk and simulator
|       ├── system                                        underlying file system, application,
and component
|       ├── test                                            Android Vendor testing framework
|       ├── toolchain                                     tool chain files
|       ├── tools                                           tools files
|       └── vendor                                        QCOM private repository
|           └── qcom
|               ├── opensource
|               └── proprietary
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh       SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh       Fastboot
flash tool
├── turbox
|   └── tools
|       ├── build_script                                    SDK build script(android build,
zip, …)
|       └── flash_tools                                     Flash tools files(qdl, …)
├── adsp_proc             aDSP image is released as a binary and no build compilation is
needed
├── aop_proc              AOP image is released as a binary and no build compilation is
needed
├── boot_images           BOOT image is released as a binary and no build compilation is
needed
├── btfm_proc             BTFM image is released as a binary and no build compilation is
needed
├── btfm_proc_CMC         BTFM_CMC image is released as a binary and no build compilation
is needed
├── cdsp_proc             cDSP image is released as a binary and no build compilation is
needed
├── modem_proc            MODEM image is released as a binary and no build compilation is
needed
├── trustzone_images      TZ image is released as a binary and no build compilation is
needed
├── venus_proc            VIDEO image is released as a binary and no build compilation is
needed
├── wdsp_proc             wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             WLAN image is released as a binary and no build compilation is
needed.
```

4.4. Download LA SDK

Follow the steps below to activate an account and download the SDK.

4.4.1. Account activation

Add the account/password information to *$HOME/.netrc* by running the following command.

```
$ echo "machine partner.thundercomm.com login <User Email> password <password>" >> ~/.netrc
```

➲ **NOTE:** Please use the email and password you have registered at *www.thundercomm.com* to replace `<User Email>` and `<password>` in the above command.

4.4.2. Download LA SDK code

Follow the steps below to download the LA SDK:

Step 1.Initialize the repository.

- In mainland China:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LA1.0 -m turbox-c610-la1.0-dev.release.es.r000002_cn.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --
no-repo-verify
```

- Out of mainland China:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LA1.0 -m turbox-c610-la1.0-dev.release.es.r000002.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --
no-repo-verify
```

Step 2.Synchronize the source code.

```
$ repo sync -c -j4 --no-tags
```

4.4.3. Update SDK version

If you have downloaded the earlier version SDK, there are two options to get the new version SDK fastly.

Option 1.Upgrade the local old folder to a new release.

Navigate to the old local code folder, you can run the `repo init` command of new release as *4.4.2.Download LA SDK code*, and then run `repo sync -j4 -c --no-tags` to receive the new code.

Option 2.Download the new release to a new folder.

Create a new folder, and run `repo init` with `--reference` value.

For example:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LA1.0 -m turbox-c610-la1.0-dev.release.es.r000002_cn.xml --
reference=<old_sdk_code> --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --no-repo-verify
$ repo sync -j4 -c --no-tags
```

➲ **NOTE**: In `--reference=<old_sdk_code path>`, the `old_sdk_code` path is the absolute path of your old sdk folder. For example:
`--reference=/home/host/turbox-c610-la1.0-dev.release.es.r000002`.

4.5. Build images

To compile code, run the following script file in the root directory of source code to build userdebug version:

```
$ ./turbox_build.sh -l -v userdebug
```

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4.6. Pack ROM package

Step 1. Run the following command to update common info for userdebug version:

```
$ ./turbox_build.sh -u --uci-imf -v userdebug
```

➲ **NOTE**: The command executes successfully with the following log displaying on shell:

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the ROM package at *turbox/output*.

- Pack **Fastboot** image

```
$ ./turbox_build.sh --zip_fastboot_build -l -v userdebug
```

- Pack **Flat** image

```
$ ./turbox_build.sh --zip_flat_build -l -v userdebug
```

➲ **NOTE**: Customers can also get ROM package provided by Thundercomm from:

- *FastbootBuild_Turbox_C610_LA1.0.R.userdebug.ES.r000002.zip*
- *FlatBuild_Turbox_C610_LA1.0.R.userdebug.ES.r000002.zip*

4.7. Flash images to board

➲ **NOTE**: If your device system is not Android, use **QDL** to flash Flat image

4.7.1. Flash Fastboot images

Step 1. Enter the **Fastboot** mode

Option 1. Enter the following command when the device in **ADB** mode:

```
$ adb root
$ adb reboot bootloader
```

Option 2. Power off the device, hold the **Volume Down (VOL-)** button and click the **Power ON** button, then release the **Volume Down (VOL-)** button about 5 seconds later.

➲ **NOTE**:

To check if the device has entered the **Fastboot** mode, run the following command:

```
$ fastboot devices
```

Successful with the following output displaying on the shell:

```
7050372c fastboot
```

Step 2. Flash images

Execute the commands below to flash images to the board.
Replace `<folder name of image package>` with the actual image package name of your product.

```
$ cd <folder name of image package>
$ bash ./turbox_flash.sh --all -r -v userdebug
```

➲ **NOTE**: Enter `./turbox_flash.sh -h` to acquire detailed usage.

```
$ ./turbox_flash.sh -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  The images except gpt,userdata,persist will be flashed if no options input
Description:
  flash images with fasboot
Options:
  -h, --help                display this help message
```

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```
  -v, --variant <debug/perf/user>      set the build variant.(default: debug)
  -f, --force                  gpt,persist and userdata will be flashed.
            User data and calibration data(BT/WIFI MAC address, etc..) will be
erased.
  -r, --reboot                 reboot the device automatically after flashing
completed.
  -b, --nonhlos                flash all bp images(xbl,rpm,ftfm,tz,dsp,modem,pmic).
  -a, --hlos                   flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp).
  --all                        flash all images
Example:
  ./turbox_flash.sh
     Flash the images except gpt,userdata,persist.
  ./turbox_flash.sh --force
     Flash all images.
  ./turbox_flash.sh --hlos
     Flash all ap images.
```

4.7.2. Flash FlatBuild images

If the open kit cannot boot up to enter the **Fastboot Mode**, flash the **FlatBuild** images using **QDL**.

Step 1. Enter **9008 Mode** (after you have finished **step 1**).

1. Press and hold the **Force_USB_Boot** key, then connect the USB device and power cable to the board.
2. Release the **Force_USB_Boot** key when the 9008 USB device has been detected.

To check if your device has entered the **9008 Mode**, run the following command:

```
$ lsusb
……
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
…….
```

Step 2. Start to flash the **FlatBuild** images.

After the images are ready, add the **QDL** tool to flash the images.
Replace **<folder name of image package>** with the actual image package name of your product.

```
$ cd <folder name of image package>
$ chmod +x qdl
$ sudo ./qdl --storage emmc prog_firehose_ddr.elf  rawprogram*.xml patch*.xml
```

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Chapter 5. New Features

Table 5-1. New features list

Item	Interface
Camera	HDMI IN
	SHDR
Audio	HDMI IN

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Chapter 6. Known Issues

- Type-A USB connector can only work under the condition that the Type-C USB connector is connected to an OTG device.
- The C410 device cannot boot up after image flash with **Fastboot**.
- The digital microphone function is not supported yet.

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